MILITARY SPECIFICATION

GENERATOR, DIRECT CURRENT: G-76B/G

This specification is approved for use by USA Labcom, Department the Army, and is available for use by all Departments and Agencies the Department of Defense.

1. SCOPE

1.1 This specification covers requirements for Generator, Direct Current, G-76B/G, a portable, foot-cranked electrical generator to provide power to military radios, coding equipment, rechargeable batteries, and other electronic equipment in the field.

2. APPLICABLE DOCUMENTS

2.1 Government Documents

2.1.1 Specifications, Standards, and Handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Military Specifications

MIL-P-116	Preservation-Packaging, Method of
MIL-P-11268	Parts, Materials and Processes Used in Electronic Communication Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-L-81352	Lacquer, Acrylic (For Naval Weapons Systems)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Labcom, Harry Diamond Labs, ATTN: SLCHD-DE-OP, 2800 Powder Mill Rd, Adelphi, MD 20783, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appear at the end of this document or by letter.

FSC 611

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Federal Standards

FED-STD-595 Colors

Military Standards

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution
MIL-STD-810C	Environmental Test Methods

2.1.2 Other Government Documents, Drawings, and Publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

Drawings

USA LABCOM Harry Diamond Laboratories (FSCM 19202)

PL 11744730	Parts List for Generator, Direct Current: G-76B/G with Bag
DL 11744730	Data List
EDL 11744730	Equipment Data List
11744814	Special Packaging Instructions

(Copies of specifications, standards, handbooks, drawings,publications and other Government documents required by suppliers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First Article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2).

3.2 Construction. Construction shall be in accordance with Data List DL11744730 and the following.

3.2.1 Materials. Materials shall be those specified by the applicable drawings.

3.2.1.1 Harmonic drive assembly. The harmonic drive assembly (11733294) shall be thoroughly cleaned and lubricated using lubricant (11733314) prior to incorporation into the electromechanical assembly (11744748).

3.2.2 Accessories. Accessories shall be required as specified in the contract (See 6.2.4)

3.2.3 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. Individual items shall not be handpicked for fit or performance. (Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.). The harmonic drives used in the G-76B/G shall be interchangeable with one another, however the components of any one harmonic drive shall not necessarily be interchangeable with the components of another harmonic drive.

3.2.4 Protective finish. Except for those parts which have been hard anodized, the surface of the unit shall be treated in accordance with MIL-F-14072, Finish No. 513.3, Final Paint Film Designation AG. Exposed aluminum surfaces shall be prepared per Finish P213.1 of MIL-F-14072. Exposed steel surfaces, such as screws, fasteners, and chains, shall be prepared per Finish P211.1 of MIL-F-14072. Exposed stainless steel surfaces shall be passivated per Specification QQ-P-35. Operating elements, such as spring loaded pins and movable arms, and visual elements, such as covers for light emitting diodes, shall be usable after applying the finish.

3.2.5 Marking. The unit shall be marked in accordance with MIL-M-13231. Panel marking shall be Group I as described in MIL-M-13231. The pin designations of the different voltages at the output connector shall be listed as a special characteristic. Both the Generator Head, Pedal and Crank Arm Assembly (11744746) and the Power Conditioning Unit (11744773) shall be marked with the unit serial number.

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3.2.5.1 Serial numbers. The serial number of the Generator, Direct Current: G-76B/G, shall be assigned in accordance with MIL-M-13231.

3.2.5.2 Visibility. Wherever practicable, components and assemblie shall be mounted so that their identification markings are readily visible with minimum disassembly of equipment.

3.3 Preconditioning.

3.3.1 Burn-in. Prior to delivery and after all processing, each Generator Head Assembly (11744747) and Power Conditioning Unit (11744773) shall be burned in (See 4.8.1).

3.3.2 Bounce preconditioning. The Generator Head Assembly (1174474 and Power Conditioning Unit (11744773) shall meet the performance requirements of 3.4 after being subjected to bounce preconditioning (See 4.8.2).

3.4 Performance requirements. The performance requirements contained herein are mandatory.

3.4.1 Rated output (resistive load). The unit (pins B and D of the output connector) shall provide a current of not less than 2.4 amperes into a resistive load of 10 ohms with an output voltage of not less than 24 volts at a cranking speed of 68 rpm; pins A and C are open circuited (See 4.8.3).

3.4.2 Output voltages. Tests to be performed at ambient temperatu unless otherwise specified.

3.4.2.1 Regulated output. The regulated output voltage (pins B and D) of the unit while cranking at 105 rpm into a 100 ohm load shall not be less than 30 volts and shall not exceed 32.0 volts DC at ambient temperature and shall not be less than 29.6 volts at -10 F nor more than 32.5 volts at 120 F (See 4.8.4).

3.4.2.2 Unregulated output. The unregulated output voltage (pins A and C) with a 10 ohm load shall be no less than 40 VDC at a cranking speed of 105 rpm (See 4.8.4).

3.4.3 Heavy load. The unit shall withstand repeated heavy loads of not less than 5.0 amperes for one minute across pins B and D, pins A and C being open circuited, with cranking speeds of 100 rpm (See 4.8.5

3.4.4 Indicators. A light emitting diode (LED) shall indicate wher the unit is providing 24 volts -- 2.4 volts (See 4.8.4). A second LED shall indicate when the unit is providing 2 amperes +- 0.2 amperes (See 4.8.4 and 4.8.5).

3.4.5 Torque test. The unit shall require a torque of 170 inch-pounds or less when cranking at 68 rpm into a resistive load of 10 ohms connected across pins B and D of the output connector; pins A and C being open circuited (See 4.8.6).

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3.4.6 Protection. The unit shall be capable of withstanding repeated momentary short circuits at one-minute intervals on the regulated line (See 4.8.7).

3.4.7 Reverse current. The unit shall not discharge the battery to which it is connected for charging; reverse current at 50 volts shall not exceed 75 microamperes on either line (See 4.8.8).

3.4.8 Acoustic Noise. The completely assembled G-76B/G shall be operated into a 50 ohm resistor and cranked by foot until 25 +- 1 VDC are measured across the resistor. The sound level measured 5 feet away from the head shall be no greater than 70 dBA (See 4.8.24).

3.5 Environmental requirements. The unit shall meet the following environmental requirements. (Units shall be unpackaged unless otherwise specified).

3.5.1 Temperature. Unit structural (3.2) and performance (3.4.1) requirements shall be met after storage at temperature extremes of -50 F and +160 F; performance requirements (3.4.1) shall be met at temperatures of -50 F and +120 F; and performance requirements (3.4.2) shall be met at temperatures of -10 F and -120 F (See 4.8.9 through (4.8.10).

3.5.2 Vibration and shock. After conditioning (See 4.8.11 through 4.8.16), the unit shall be capable of set-up to the high and low profile operating mode such that an operator can sit on the seat and manually crank the unit at 68 +- 10 rpm. The unit shall also meet the requirements of para. 3.4.1, 3.4.2 and 3.4.4.

3.5.3 Humidity. Unit structural (3.2) and performance (3.4) requirements shall be met after prolonged exposure to humidity conditions encountered in tropical areas (See 4.8.17).

3.5.4 Immersion. The unit shall meet performance requirements (3.4) after being immersed in water. There shall be no leakage within the Generator Head or PCU (See 4.8.18).

3.5.5 Salt atmosphere. Unit structural (3.2) and performance (3.4) requirements shall be met after exposure of the unpackaged unit to salt atmospheres as may be encountered in coastal areas (See 4.8.19).

3.5.6 Fungus. Unit structural (3.2) and performance (3.4) requirements shall be met after prolonged exposure of the unpackaged unit to conditions favorable to fungus attack, as may be encountered in tropical climates (See 4.8.20). Examination after exposure shall show no evidence of fungus.

3.5.7 Blowing sand and dust. The unit shall meet performance requirements (3.4) after being subjected to blowing sand and dust as encountered in the desert (See 4.8.21).

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3.6 Electromagnetic compatibility. The unit shall comply with the radiated and conducted emission requirements of MIL-STD-461, Notice 4, as specified below:

RE 02
RE 02.1
CE 04

(See 4.8.22).

3.7 Reliability. The unit shall have a mean time between failure (MTBF) of 750 hours minimum acceptable value (MAV) and a best operation characteristic (BOC) of 2250 hours (a discrimination ratio of 3). The requirement applies under the following conditions:

a. Temperature min of 104 F or max of 145 F as measured in the generator head.
b. Continuous operation with air cooling of generator bearings and regulator box.
c. Unless otherwise specified, the environment will be fixed ground as defined in MIL-STD-781.

(See 4.8.23).

3.8 Workmanship. The equipment shall be manufactured and assembled with the applicable portion of the following paragraphs of MIL-P-11268.

a. General requirements for plastic material and plastic parts.

b. Wiring and cabling, including: slack, protection, clearance, splicing, connections in general, grounding, shielding on wire and cable.

c. Flux and cleaning agents for solder.

d. Process for solder.

e. Cleaning

f. Riveting

g. General requirements for securing parts.

h. Welding

i. Mechanical operation of controls

j. Self-tapping screws

Requirements of MIL-STD-454, Requirement 9, shall be used to establish acceptable workmanship criteria.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all required inspections as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspections specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspection shall be established by the contractor and approved by the Government prior to use. Gage and instrument settings shall be a function of the equipment and independent of the operator. Test equipment shall be marked to indicate the limit of acceptance for each test. Test equipment accuracy shall be taken into account in setting the acceptance limits. Special test equipment required for acceptance tests is listed in EDL 11744730.

4.1.2 Standard conditions. Tests shall be conducted at standard ambient conditions of temperature, pressure, and humidity as defined in MIL-STD-810C, unless otherwise specified. The following tolerances on test equipment accuracy also apply, unless otherwise specified:

a.	Time	+- 2 percent
b.	Resistance, current	+- 2 percent
c.	Frequency, period	+- 2 percent
d.	Vibration amplitude	+- 10 percent
e.	Force, weight, or torque	+- 2 percent
f.	Voltage	+- 0.5 percent
g.	Acoustic Noise Level	+- 1.0 dBA

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (does not include packaging)(See 4.3)

b. Quality conformance inspection.

(1) Quality conformance inspection of equipment before packaging. (See 4.4)

(2) Quality conformance inspection of packaging (See 4.9)

c. Inspection covered by subsidiary documents. (See 4.5)

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4.3 First article inspection. The first article inspection shall consist of the First Article Sample Inspection (Table II), the Classification of Defects (4.6), and Material and Component Certifications (4.7). In addition, variables measurements shall be made of the ten sets of parts listed in 6.4. The parts shall be among those used in the construction of the First Article sample. The measurements shall be compared with the drawings, and the comparison shall be submitted with the first article sample test data. In addition, the data on the ten sets of parts shall be certified by an independent laboratory approved by the Government.

Table II. First Article Sample Inspection

Inspections	Req Para	Test Para	Unit Number
Inspection Covered by Subsidiary Documents	-	4.5	All
All Group A tests	See Table III		All
Group B tests*	See Table IV		All*
High Temperature	3.5.1	4.8.9	1,2
Low Temperature	3.5.1	4.8.10	1,2
Vibration	3.5.2	4.8.11	1,2
Vibration-Temperature	3.5.2	4.8.12	3,4
Shock (Drop)	3.5.2	4.8.13	1,2
Shock (Drop-Temperature)	3.5.2	4.8.14	3,4
Bounce	3.5.2	4.8.15	1,2
Shock (Bench)	3.5.2	4.8.16	1,2
Humidity	3.5.3	4.8.17	5
Immersion	3.5.4	4.8.18	6
Salt fog	3.5.5	4.8.19	7
Fungus	3.5.6	4.8.20	8
Blowing Sand and Dust	3.5.7	4.8.21	9
Electromagnetic compat	3.6	4.8.22	10
Reliability	3.7	4.8.23	11 - 20

* All units shall be subjected to all Group B tests except the temperature tests, which will be conducted as indicated in this table; AQL's in table IV do not apply

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4.3.1 Reinstitution of tests. Acceptance of the FAS does not relieve the contractor from meeting all requirements of this specification throughout the contract. The Government reserves the right to independently verify all requirements by repeating FAS inspections or any portion thereof on any production lot. If such verification indicates failure to meet requirements previously verified by FAS tests, the contractor shall be required to take corrective action. If the requirements are not met on the first lot to which the correction can be applied, the Government reserves the right to institute tightened inspection on a lot basis, by adding to the normal quality conformance inspections all or such parts of the FAS tests needed to verify corrective action. Normal quality conformance inspection will be resumed after two consecutive lots have satisfactorily passed the tightened conditions (see 6.5).

4.4 Quality conformance inspection of equipment before packaging. Quality conformance inspection shall include inspection by subsidiary documents (4.5), classification of defects (4.6), material and component certifications (4.7), Group A Tests (Table III), and Group B Tests (Table IV). The contractor shall notify the Government QA representative prior to proceeding with inspection for Classification of Defects and Group A and B tests. Inspection may be made at a higher assembly level than specified provided that equal quality assurance provisions are satisfied but not at a lower level than specified. If a unit should fail any Group B test, testing shall be stopped and data and analysis shall be forwarded to the technical agency within two working days. The technical agency will evaluate the analysis and recommend changes where indicated.

4.4.1 Lot formation. A lot shall consist of units produced at one manufacturing location in accordance with stabilized methods and techniques, the same drawing, and specification, and revisions (except for those introduced without obsolescence), under continuous production. One lot shall comprise one month's production but not less than 50 units.

4.4.2 Sampling. Sampling shall be in accordance with MIL-STD-105 and this document. Unless otherwise specified, inspection level II shall apply.

4.4.3 Data recording. A record shall be made of the results of each performance test but only the defect summary need be recorded in the inspection of the Classification of defects. The data shall be fowarded to the technical agency within two working days of the Government acceptance of the item or lot. If inspections are conducted on a sublot basis, the acceptance data for each sublot shall be so identified.

4.4.4 Disposition of lot samples. After passing the prescribed tests, the samples shall be refurbished as necessary. One sample shall be selected as the comparison sample (see 6.5). The remaining samples shall be returned to the lot.

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Table III. Group A Tests

Inspections	Reqt Para	Test Para	Test Equip	Acceptance Criteria
Burn in	3.3.1	4.8.1	SIE	100% screening
Bounce preconditioning	3.3.2	4.8.2	STE	100% screening
Rated output	3.4.1	4.8.3	SIE	100% screening
Output voltage	3.4.2	4.8.4	SIE	100% screening
Heavy load	3.4.3	4.8.5	SIE	100% screening
Indicators, 24 volt	3.4.4	4.8.4	SIE	100% screening
Indicators, 2 amperes	3.4.4	4.8.5	SIE	100% screening
Torque	3.4.5	4.8.6	SIE	100% screening
Acoustic Noise	3.4.8	4.8.24	SIE	100% screening

Table IV. Group B Tests

Inspections	Reqt Para	Test Para	Test Equip	Acceptance Criteria
Protection	3.4.6	4.8.7	STE	AQL 1.0
Reverse current	3.4.7	4.8.8	STE	AQL 1.0
High temperature	3.5.1	4.8.9	STE	AQL 1.0
Low temperature	3.5.1	4.8.10	STE	AQL 1.0

4.5 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification and the inspection requirements specified in the contract or purchase order.

Item	Where Required
Construction	3.2
Materials	3.2.1
Accessories	3.2.2
Dimensions	3.2.3.1
Finish	3.2.4
Marking	3.2.5
Workmanship	3.8
Preservation-Packaging	5

4.6 Classification of defects. The assembly shall be inspected by using MIL-STD-252 and the following Classification of Defects and applicable drawings.

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4.6.1 Foot crank generator assembly (11744731)

Category	Defect	Method	Acceptance Criteria
Major			
101	Running torque	SME	100% Inspection
102	Acoustic Noise	DS-5	100% Inspection
Minor			
201	Torque on all fasteners	SME	AQL 1.0
202	Function of hinges	Manual	AQL 1.0
203	All weld joints	Visual / SME	100% inspection
204	Finish	Visual	100% inspection

4.6.1.1 Seat assembly (11744732)

Category	Defect	Method	Acceptance Criteria
Major			
101	Proper pivot of seat	Visual	100% inspection
102	Proper fit of release pin	Visual	100% inspection
Minor			
201	Proper torque on all fastners	SME	AQL 1.0

4.6.1.2 Power module assembly (11744734)

Category	Defect	Method	Acceptance Criteria
Major			
101	Operation of plunger	Visual	100% inspection
Minor			
201	Proper operation of low profile support bracket	Visual	100% inspection

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4.6.1.3 Generator head, pedal and crank arm assembly (11744746)

Category	Defect	Method	Acceptance Criteria
Major			
101	Proper operation of quick release pins	Visual	100% inspection
102	Proper operation of pedals	Visual	100% inspection
103	Fit of crank arm adapter	Visual	100% inspection
Minor			
201	Proper torque on all fastners	SME	AQL 1.0

4.6.1.4 Generator head assembly (11744747)

Category	Defect	Method	Acceptance Criteria
Major			
101	Water seal	DS-1	2 samples/lot ac 0/re 1
Minor			
201	Conformal adhesive	Visual	100% Inspection
202	Torque on screws	SME	AQL 1.0

4.6.1.5 Electromechanical assembly (11744748)

Category	Defect	Method	Acceptance Criteria
Minor			
201	Adhesive or screws	Visual/SME	AQL 1.0
202	Proper position of low speed shaft bearings	DS-2	100% Inspection
203	Cleanliness of harmonic drive	Visual	100% Inspection
204	Lubrication of harmonic drive	Visual (11734028)	100% Inspection

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4.6.1.6 Alternator cup and high speed shaft assembly (1174474S

Category	Defect	Method	Acceptance Criteria
Minor			
201	Proper weight, amount, and distribution of grease	11734028 Visual	100% Inspec
202	Torque on screws	SME	AQL 1.0

4.6.1.7 High speed shaft assembly (11744750)

Category	Defect	Method	Acceptance Criteria
Major			
101	Crimp rollover	DS-3	100% Inspec
102	Proper bearing lubrication	Visual (minimum magnification of 7X)	100% Inspec
103	Proper assembly of shaft bearing and rotor	Visual (minimum magnification of 7X)	100% Inspec
104	No damage to shaft, bearing or rotor	Visual (minimum magnification of 7X)	100% Inspec
Minor			
201	Torque on screws	SME	AQL 1.0

4.6.1.8 High speed shaft (11744753)

Category	Defect	Method	Acceptance Criteria
Minor			
201	Proper interference fit	SME	100% Inspec
202	Proper fit of woodruff key	Visual	100% Inspec

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4.6.1.9 Power conditioning unit (11744773)

Category	Defect	Method	Acceptance Criteria
Major			
101	Solder seal on plate screws	Visual	100% Inspection
102	Seal provided by plate and connector gaskets (water seal)	DS-1	2 samples/lot ac 0 re 1
Minor			
201	Heat sink and thermal adhesive	Visual	100% Inspection
202	Interchangeability of power output connector	DS-4	2 samples/lot ac 0/re 1

4.6.1.10 Board assembly A (11733389)

Category	Defect	Method	Acceptance Criteria
Major			
101	Electrical output	STE	100% Inspection
102	Torque on heat sink screw	SME	100% Inspection
103	Part inventory	Visual	100% Inspection

4.6.1.11 Board assembly B (11733411)

Category	Defect	Method	Acceptance Criteria
Major			
101	Electrical output	STE	100% Inspection
102	Torque on heat sink screw	SME	100% Inspection
103	Part inventory	Visual	100% Inspection

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4.6.1.12 LED support assembly (11733419)

Category	Defect	Method	Acceptance Criteria
Major			
101	Electrical function	STE	100% Inspection

4.6.1.13 Tripod assembly (11744786)

Category	Defect	Method	Acceptance Criteria
Major			
101	Proper fit of release pins	Visual	100% Inspection
102	Proper movement of hinged parts	Visual	100% Inspection
103	Proper extension & locking of legs	Visual	100% Inspection
Minor			
201	Tightness of sash chain	Visual	100% Inspection
202	Proper torque on fastners	SME	AQL 1.0

4.6.1.14 Front leg assembly (11744830)

Category	Defect	Method	Acceptance Criteria
Major			
101	Weld on key	X-ray	100% Inspection

4.6.1.15 Left and right rear leg assemblies (11744831)

Category	Defect	Method	Acceptance Criteria
Major			
101	Proper operation of hand Knob	Visual	100% Inspection

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4.6.2 Defect standard

DS-1 Immerse in accordance with MIL-STD-810C, Method 512.1, Procedure 1. Any evidence of moisture on disassembly shall constitute a failure.

DS-2 To check low speed shaft, fit bearing on both ends. Place TRUARC in shaft groove, place bearings in fully seated position, measure distance from outside to outside of bearing. This distance shall be as indicated on the drawing.

DS-3 Hold high speed shaft stationary, apply 50 inch-pounds to torque to bearing inner race. The bearing shall not move on the shaft.

DS-4 To check the quality of the assembly, connector (HDL 11738229) shall be inserted and removed from the power output connector of the Power Conditioning Unit with no undue effort.

DS-5 For the method of performing the acoustic noise test, refer to para 4.8.24

4.7 Material and component certification. Prior to acceptance under this procurement, the contractor shall demonstrate by means of certifications or statements of findings that only materials and components conforming to the item specification, drawings, and subsidiary documents (4.5) have been used.

4.8 Test methods. (See 4.1.1)

4.8.1 Burn-in. The generator head assembly (11744747) with a power conditioning unit (11744773) attached shall be connected to an electric motor and be operated over a 3 1/2 hour period (1/2 hour on, one hour off, 1/2 hour on, 1 hour off, 1/2 hour on) at a speed of 68 rpm into a 10 ohm load.

4.8.2 Bounce preconditioning. The generator head assembly (11744747) with a power conditioning unit (11744773) attached shall be connected together and mounted on a suitable fixture and then placed on the table of a package tester (type 1000 SC, L.A.B. Corporation, Skaneateles, NY, or equivalent). The package tester, shafts in phase, shall have a speed such that the items bounce 1/4 inch from the bed. The items shall be subjected to this preconditioning for 10 minutes. After the bounce preconditioning, the unit shall not be repaired, aligned, cleaned or otherwise changed prior to being subjected to performance requirements.

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4.8.3 Rated output. The generator head assembly (11744747) with a power conditioning unit (11744773) attached (pins B and D) shall be connected to a resistance load of 10.0 ohms and driven at 68 rpm. The output current shall be measured with an external ammeter and shall meet the output requirements of 3.4.1.

4.8.4 Output voltage. The generator head assembly (11744747) with a power conditioning unit (11744773) attached shall be driven at 105 rpm into a 100 ohm load. The regulated output voltage shall meet the requirements of 3.4.2.1. The accuracy of the voltage indicating LED shall be determined by comparison with an external voltmeter. Driving the generator head assembly and power conditioning unit at 68 to 100 rpm into a 10 ohm load, the accuracy of the LED shall meet the requirements of 3.4.4. The voltage output of the unregulated line shall be measured while cranking at 105 rpm into a 10 ohm load and shall meet the unregulated requirements of 3.4.2.2.

4.8.5 Heavy load. The generator head assembly (11744747) with a power conditioning unit (11744773) attached shall be driven at 100 rpm, and the resistance load (across pins B and D) shall be adjusted to obtain an output current of at least 5.0 amperes. The generator assembly shall be so driven 10 times for 1.0 minute periods with 5 minute intervals between tests during which time the generator assembly shall not be cranked at all. The generator and conditioning unit shall meet the rated output requirements (3.4.1) after this test. The accuracy of the 2.0 ampere indicating LED shall be determined by comparison with an external ammeter and shall meet requirements of 3.4.4

4.8.6 Torque. The mechanical torque applied to the generator assembly shall be measured. The unit shall meet the requirements of 3.4.5.

4.8.7 Protection. The generator head assembly (11744747) with a power conditioning unit (11744773) attached shall be driven at 68 rpm, and a short circuit shall be momentarily applied across the regulated voltage terminals. The output shall be short circuited a minimum of five times with one minute intervals between short circuits. During the short circuit, the cranking speed shall be maintained. After the test, the generator head and power conditioning unit shall meet the requirements of 3.4.2.1 and 3.4.6.

4.8.8 Reverse current. A DC power supply with an adjustable output voltage from 0 to 50 VDC shall be connected to the output terminals on both circuits of the generator assembly conditioning unit in correct polarity with the generator not being cranked. The voltage of the power supply shall be gradually increased to 50 VDC and the requirements of 3.4.7 shall be met. The generator and conditioning unit shall meet the rated output requirements (3.4.1) after this test.

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4.8.9 High temperature. The unit shall be temperature conditioned in accordance with MIL-STD-810C, method 501.1, Procedure II (See 3.5.1) At least once during each 12 hour cycle after the unit has stabilized a 120 F, once at 120 F after the third cycle, and once at ambient temperature upon completion of the test, the unit shall undergo the rated output test (4.8.3) and the output voltage test (4.8.4). The unit shall meet requirements (3.4.1 and 3.4.2).

4.8.10 Low temperature. The unit shall be temperature conditioned in accordance with MIL-STD-810C, Method 502.1, Procedure I (See 3.5.1). At least once during the -50 F portion of the test, after the unit has stabilized, and once at ambient temperature upon completion of the test, the unit shall undergo the rated output test (4.8.3). The unit shall meet the rated output voltage requirement (3.4.1). At least once during the -10 F portion of the test, while cranking into a 10 ohm load, the unit shall undergo the output voltage test (4.8.4). After cranking for 4.5 +- 0.5 minutes, the unit shall meet the requirements of 3.4.2

4.8.11 Vibration test. The unit shall be tested in accordance with MIL-STD-810C, Method 514.2, Procedure X, curve AX (See 3.5.2). The generator head assembly (11744747) and power conditioning unit (11744773) shall be tested and shall meet the requirements of 3.4.1 and 3.4.2.

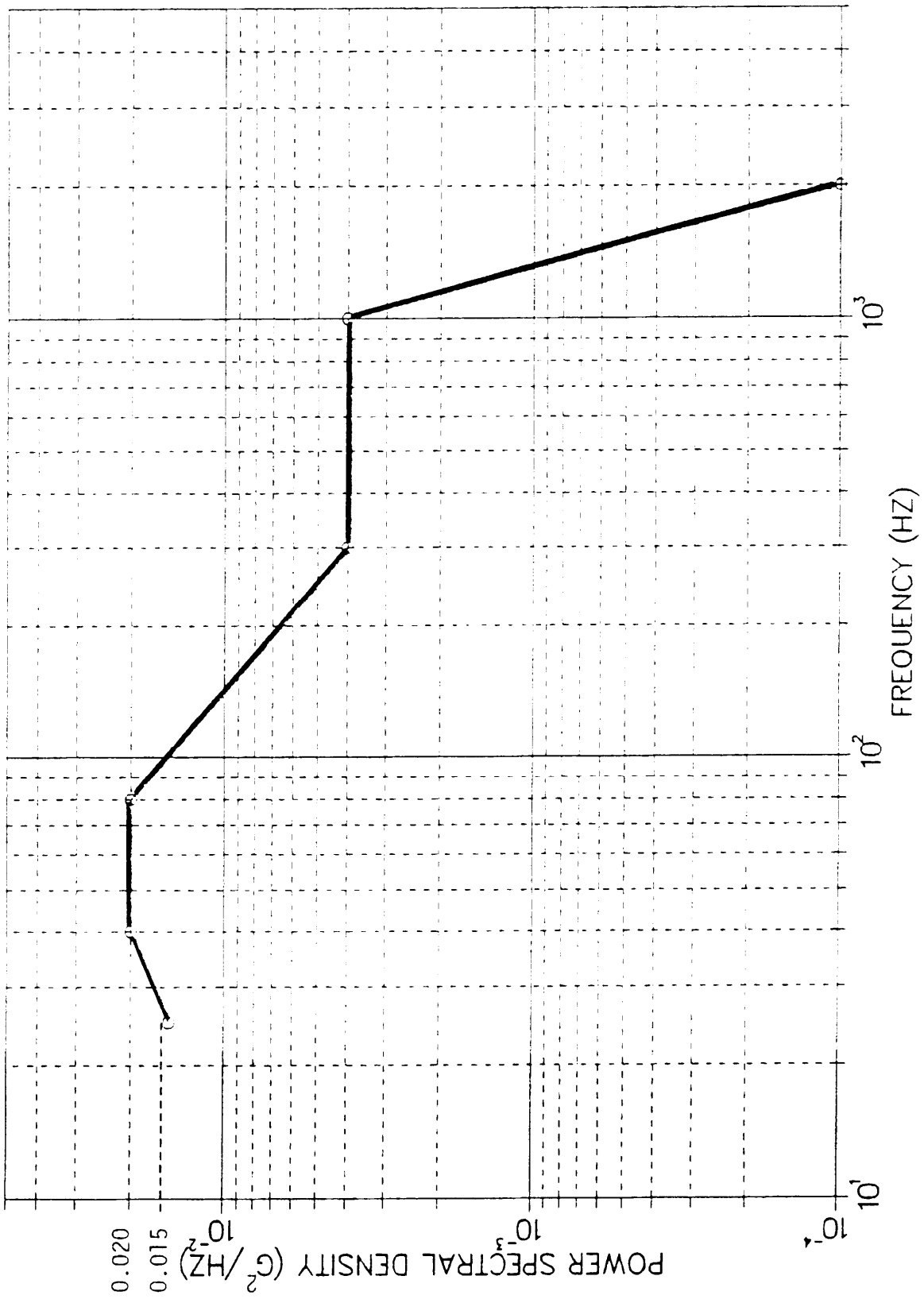
4.8.12 Vibration-temperature test. The unit in its combat carrying bag shall be tied down to the vibration fixture in accordance with the normally prescribed tie down procedures so that vibration is applied through the units vertical axis. The unit shall be at a temperature of 73 F. The unit shall be vibrated for two hours in accordance with the random vibration spectrum shown in figure 1. The inputs specified are those associated with the fixture and not the unit. The control accelerometer shall be on the fixture. Input tolerances for ramdom vibration are to be held to +- 3 dB up to 200 Hz and +- 5 dE over 200 Hz. At completion, the unit shall be subjected to the following sinusoidal vibration environment:

Frequency (Hz)	Vibration Amplitude (g)	Sweep Rate	Total Time/Axis (min)
10 - 50	1.5	10-150 Hz in 5 min	5
50 - 150	2.5		

The entire procedure described above shall be repeated two more times for a total of three cycles, first with the vibration imparted through

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Figure 1. Surface and Air Transportation Vibration



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the unit's longitudinal axis, and second with the vibration imparted through the unit's transverse axis. At the completion of each of the three cycles, the unit shall meet the requirements of paragraphs 3.5.2 and 3.4.1, and the combat carrying case shall meet the requirements of 3.5.2. The entire procedure described above shall be repeated two more times, once at a temperature of +130 +- 2.5 F and again at a temperature of -30 +- 2.5 F. These temperature extremes are to be achieved at a rate of 25 degrees per hour, and the unit shall be stabilized at the test temperature for at least 16 hours before vibration. The control temperature sensor shall be attached to the skin of the container. During changes in vibration axes, the unit test temperature shall be maintained. The unit shall be returned to ambient temperature before the functional tests and appropriate inspections are conducted. At the completion of the high temperature testing, the unit shall meet the requirements of paragraphs 3.5.2 and 3.4.1; at the completion of the low temperature testing, the unit shall meet the requirements of 3.5.2 and 3.4.1.

4.8.13 Shock (Transit Drop) test. The unit shall be conditioned in accordance with MIL-STD-810C, Method 516.2, Procedure II (See 3.5.2). After conditioning, the unit shall meet requirements of 3.4.1 and 3.5.2.

4.8.14 Shock (Drop-Temperature) test. The unit in its combat carrying bag shall be placed inside a conditioning chamber at ambient conditions. The chamber temperature shall be increased at the rate of 25 degrees per hour and stabilized at +130 +- 2.5 F for a period of not less than 16 hours. The unit shall be removed from the conditioning chamber and raised to a height such that its lowest corner is 30 inches above a concrete impact surface. The generator shall be the impacting part of the unit. Within five minutes after removal from the conditioning, the unit shall be released using a quick-release mechanism The unit shall be oriented at release such that its center of gravity is directly above the striking corner. After visually inspecting the unit and combat carrying bag for damage, it shall be returned to the conditioning chamber and reconditioned at +130 +- 2.5 F for a period of not less than 1 hour and dropped on a different corner and inspected. This procedure will be followed until the unit has been dropped on all four of the generator's corners. At the completion of the four drops and inspections, the unit shall meet the requirements of paragraph 3.4.1 and 3.5.2. Using a second unit and combat carrying bag, the entire procedure described above shall be repeated but at a conditioning temperature of -30 +- 2.5 F.

4.8.15 Bounce (loose cargo) test. The packaged unit shall be tested in accordance with MIL-STD-810C, Method 514.2, Procedure XI, Part 2 (See 3.5.2). After conditioning, the unit shall meet requirements of 3.4.1 and 3.5.2.

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4.8.16 Shock (bench handling) test. The unit shall be tested in accordance with MIL-STD-810C, Method 516.2, Procedure V (See 3.5.2). The unit shall meet requirements of 3.4.1 and 3.5.2 after conditioni

4.8.17 Humidity test. The unit shall be tested in accordance wit MIL-STD-810C, Method 507.1, Procedure III and shall meet the require of 3.4.1 and 3.5.3 during and after conditioning.

4.8.18 Immersion test. The unit shall be tested in accordance wi MIL-STD-810C, Method 512.1, Procedure I. The equipment shall meet t requirements of 3.4 after conditioning before opening the generator head or power conditioning unit. There shall be no evidence of leak after opening these items.

4.8.19 Salt fog test. The unit shall be tested in accordance wit MIL-STD-810C, Method 509.1, Procedure I. After conditioning, visual inspection shall show no flaked, pitted, blistered, or otherwise loosened finish on metal surface. The unit shall meet the requireme of 3.4.1.

4.8.20 Fungus test. The unit shall be tested in accordance with MIL-STD-810C, Method 508.1. On completion of conditioning, the unit shall be inspected in accordance with Procedure I, and shall meet th requirements of 3.5.6 and 3.4.1.

4.8.21 Blowing sand and dust test. The unit shall be subjected t tests in accordance with Method 510.1, Procedure I, of MIL-STD-810C shall meet the requirements of 3.4.1 and 3.5.7 after the test. Dust in any component shall be cause for failure.

4.8.22 Electromagnetic compatibility. The unit shall be subjecte to an electromagnetic interference test in accordance with MIL-STD-462, Notice 4, Methods RE 02, RE 02.1 and CE 04 and shall meet the requirements of 3.6.

4.8.23 Reliability test. The reliability test shall demonstrate the MTBF required in paragraph 3.7. The test shall be run according to the conditions of MIL-STD-781, test plan XXI-C. The generator he assembly (11744747) and power conditioning unit (11744773) shall be connected to a resistance load of 10.0 ohms and cranked at a speed o 68 rpm. A failure for reliability purposes shall be defined as the failure of any part of the unit to meet the requirements of this specification. Each unit shall be continuously monitored to detect failures. Each unit shall meet the requirements of 3.4.1 upon completion of the test.

4.8.24 Acoustic noise. A sound meter, GENRAD 1982 Precision Soun Meter and Analyzer, or equivalent, shall be placed so that the tip o the microphone is directly in front of the generator head, on a leve with it and at a distance of five feet from it. The long axis of th meter shall be at an angle of 20 degrees to the line joining the

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generator head and the tip of the microphone. The d-c voltage output from the generator's power conditioning unit shall be connected to a 50 ohm resistor and the voltage read with an appropriate meter. The generator shall be operated by foot in a location having a background noise level of 55 dBA or less. Sound level readings shall be made when the voltage output under the conditions described is 25 +- 1 volts. Acoustic measurements shall be made on the dBA scale of the sound level meter. The unit shall meet the requirements of 3.4.6.

4.9 Quality conformance inspection of packaging. Packaging shall be inspected as provided by 11744814 to determine conformance with the requirements.

4.9.1 Rough handling test (packaging). When the rough handling test in accordance with MIL-P-116 is required by contract the following functional test shall be conducted to determine freedom from operational malfunction caused by rough handling:

Requirement	Method	Cranking Speed
3.4.1	4.8.4	68 rpm

5. PRESERVATION-PACKAGING

5.1 Preservation-packing. Preservation and packing shall be in accordance with 11744814.

5.2 Shipping containers. Shipping containers shall be as specified in 11744814.

5.3 Marking.

5.3.1 Military. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with 3.2.5 and MIL-STD-129.

6. NOTES

6.1 Intended use. The intended use of the Foot-cranked generator is to provide power to operate radios in forward areas, to operate electronic fuze setters and to recharge batteries.

6.2 Ordering information

6.2.1 Documents. Procurement documents should specify the title, date, and number of this specification.

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6.2.2 Data Requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirement identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD Far supplement, Part 27, Subpart 2.7.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

Para. No.	Data Requirement Title	Appl DID No.
4.1	Inspection System Program Plan	DI-R-4803
4.3	First Article Test Plan	DI-T-53154
4.3	First Article Inspection Report	DI-T-4702
4.4	Lot Summary Inspection Record	DI-R-1757
4.3.4.4	Test Data Recording Format for Magnetic Tape	DI-R-10579
4.1.1	Quality Engineering Acceptance Inspection Equipment Design Concepts	DI-R-1713
4.4	Failure Analysis and Corrective Action Report	DI-R-5299C
4.7	Certificates of Compliance	DI-E-2121

6.2.3 First Article. When a first article inspection is required, the items inspected shall be a first article sample. The first article sample should consist of twenty units (Foot Crank Generator Assembly-11744731). The contracting officer should include specific instruction in acquisition documents regarding arrangements for examinations, approval of first article test results and dispositions of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.2.3.1 Lead time. Government approval of the first article will require a maximum of twenty working days. If there are failures, additional time will be required.

6.2.3.2 Disposition of First Article Samples. After inspection, the first article sample may be submitted with a subsequent lot after being washed, cleaned, dried, and refurbished as necessary. The units will be subjected to quality conformance tests for that lot.

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6.2.4 Accessories. Accessories that may be ordered include the following:

11738242 Accessory Kit, Field Radio Equipment, MK-2120/G

11738243 Accessory Kit, Permissive Action Link and Fuze Setter, MK-2121/G

6.3 Engineering information. Upon request, copies of the data collected in the acceptance of a lot will be made available to the Government representative. Additional data or data requiring special requirements, when needed for engineering purposes, will be specifically defined by the contract.

6.4 First article sample parts list. Ten sets of the following parts shall be compared with drawings as specified.

Drawing No.	Title	Characteristic
11733353	Rotor Assembly	All physical
11733373	Core Rotor	All physical
11733374	Magnet	All magnetic
11733375	Sleeve, Rotor	All physical
11744733	Rod Assembly (Seat)	All physical
11744741	Seat Clamp	All physical
11744742	Part D (Seat Assembly)	All physical
11744743	Pivot (Seat Assembly)	All physical
11744752	Cup, Alternator	All physical
11744753	Shaft, High Speed	All physical
11744754	Shaft, Low Speed	All physical
11744755	Crank Arm Mating Piece	All physical
11744761	Housing-Machined	All physical
11744771	Slide Block	All physical
11744787	Yoke, Stand	All physical
11744804	Lamination Stack	All physical
11744810	Mounting Bracket Assembly	All Physical

6.5 Comparison Sample. The contract should provide for delivery of one unit from each lot sample, refurbished as necessary, within two days of lot acceptance to the technical agency for use in comparison sample tests (see 4.3). These samples are subject to periodic retesting by the Government to insure that first article requirements continue to be met. Failure of these samples will not affect acceptance of the lot, but may indicate future mandatory action regarding quality conformance tests.

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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